PROJECT BASED LEARNING (PBL) Synopsis
AI & ML Lab Using Python

Credit Card Fraud Detection Using Machine Learning

Problem Id: 12

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PROBLEM STATEMENT:

Credit card fraud is a major financial issue worldwide, causing significant losses to banks and customers. Detecting fraudulent transactions is challenging due to the highly imbalanced nature of the dataset, where fraudulent transactions are very rare compared to legitimate ones.

PROBLEM DESCRIPTION:

Credit card fraud is a serious issue in the financial sector, leading to huge monetary losses for banks and customers. With the increasing use of online transactions, detecting fraudulent activities has become more challenging. Fraudulent transactions are very rare compared to legitimate ones, making the dataset highly imbalanced. Traditional rule-based systems fail to detect new and evolving fraud patterns effectively.

This project aims to build a machine learning-based classification system that can automatically detect whether a transaction is legitimate or fraudulent. By analyzing transaction-level data such as amount, time, and other features, the system can help financial institutions take quick actions and reduce financial risks.

STEPS OF IMPLEMENTATION:

1. Dataset Selection

   • A publicly available credit card fraud detection dataset will be used.

2. Data Preprocessing

   • Handling missing values

   • Removing duplicate records

   • Feature scaling and normalization

   • Data transformation for consistency

3. Handling Imbalanced Data

   • Techniques such as oversampling (SMOTE) or undersampling will be applied to balance the dataset.

4. Feature Selection

   • Important features will be selected to improve model performance and reduce complexity.

5. Model Training

   The following machine learning algorithms will be implemented:

   • Logistic Regression

   • K-Nearest Neighbours (KNN)

   • Support Vector Machine (SVM)

6. Model Evaluation

   Models will be evaluated using:

   • Accuracy

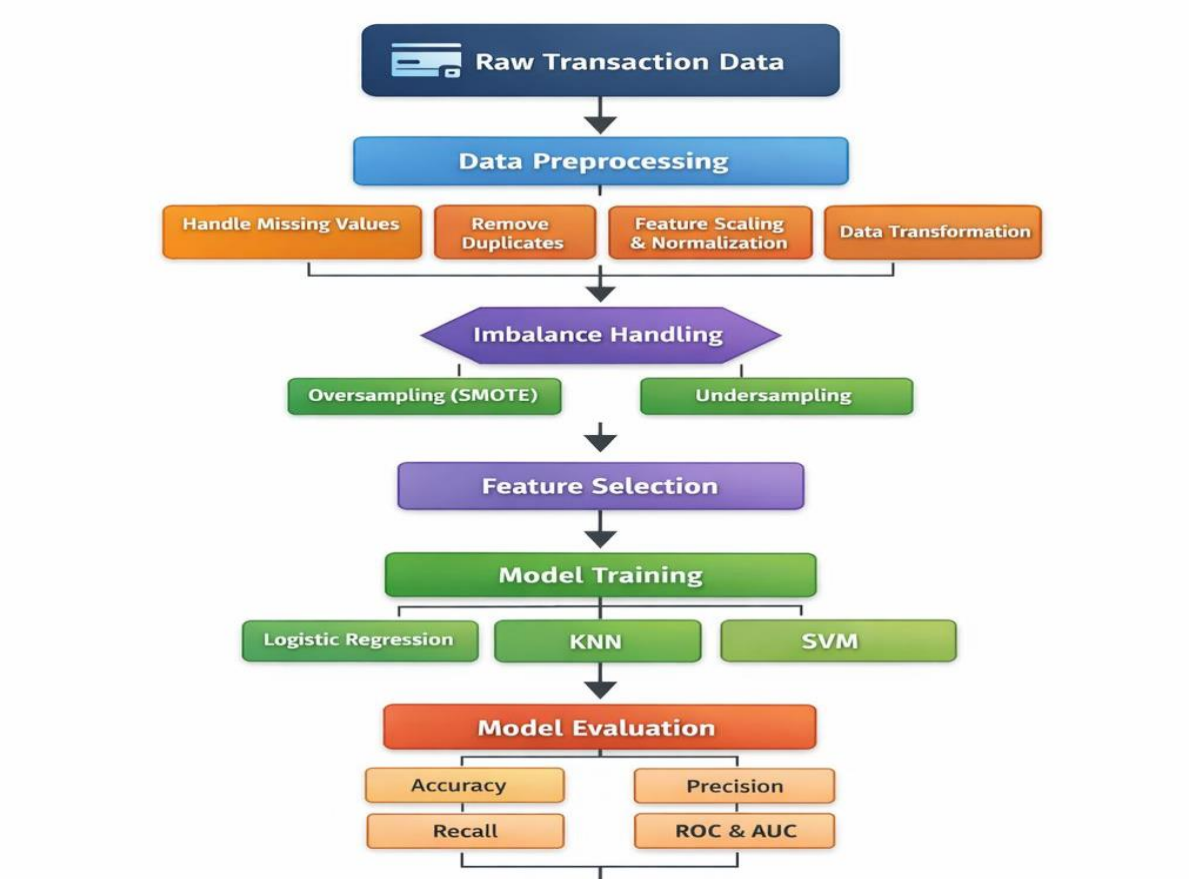
   • Precision

   • Recall

   • F1-score

   • Confusion Matrix

   • ROC Curve and AUC Score



EXPECTED MODULES/LIBRARIES TO BE USED :

NumPy: Used for numerical operations and array handling.

Pandas: Used for data manipulation and preprocessing.

Matplotlib: Used for plotting graphs and visualizations.

Seaborn: Used for advanced statistical visualizations.

Scikit-learn: Used for machine learning models and evaluation metrics.

Imbalanced-learn (SMOTE): Used for handling imbalanced datasets.

EXPECTED EVALUATION METRICS AND PLOTS TO BE USED :

Accuracy: Measures overall correctness of the model.

Precision: Important to reduce false fraud predictions.

Recall: Crucial to detect maximum fraudulent transactions.

F1-Score: Balances precision and recall.

Confusion Matrix: Shows correct and incorrect classifications.

ROC Curve & AUC Score: Evaluates model performance across thresholds.

REQUIRED TOPICS FROM THE SUBJECT:

Data Structures (Arrays/Lists): Used to store and process transaction data efficiently.

Functions: Help in modularizing code for preprocessing, training, and evaluation.

File Handling: Used to read datasets (CSV files) and store outputs.

Libraries & Modules: Enable use of pre-built functions for ML and visualization.

Basic Python Concepts: Includes loops, conditions, and data types for logic building

PLATFORM REQUIRED:

Jupyter Notebook / Google Colab / Python IDE (VS Code)

BOOKS AND LINK SOURCES:

Python Official Documentation – https://docs.python.org

Scikit-learn Documentation – https://scikit-learn.org

GeeksforGeeks (Python & ML Concepts) – https://www.geeksforgeeks.org

Artificial Intelligence: A Modern Approach – Stuart Russell & Peter Norvig